# Which Way Does Fairness Go? The Selection Rate Predicts Whether an Attribute-Blind Constraint Gives or Takes


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***Abstract*—A group-fairness constraint is a statement about a gap, and says nothing about which of two very different worlds should close it: one where the disadvantaged group is lifted, and one where the advantaged group is pulled down. Both satisfy the constraint and both score identically on the metric that certifies them. We ask which of the two occurs, and show it is predictable in advance from a quantity every deploying organisation already has — the rate at which its current model says yes. Across 18 independent populations spanning two household surveys, US mortgage records, criminal-justice risk data, US law-school outcomes and the Dutch national census, a parity constraint withdraws favourable decisions below a crossover and extends them above it. On UCI Adult, six in-processing methods each shrink the pool by 7.9–22.1%; on mortgage lending the identical constraint grows it. The certifying metric reports the same success in both cases. The relationship is not an artifact of how the selection rate is varied: holding the task, features, groups and fitted scores fixed and moving only the decision threshold reproduces both the direction and the location of the transition. It survives a twenty-five-fold range of constraint tolerance, a boosted-tree learner, and a protected attribute swapped from sex to race. Crucially, it is bounded: under *post-processing*, which reads the protected attribute at decision time, the effect vanishes entirely ($r = -0.024$ against $+0.585$), and contemporaneous theory says why — in that regime the direction is determined, not distributional. The theory's prediction for that regime holds in 18 of 18 populations post-hoc and 9 of 9 in a pre-registered replication. The practical result is a procedure: locate your own crossover by sweeping your deployed model's threshold, needing no new data. We also report a *sealed* prediction on nine populations never previously measured, committed before the runs, which failed at four of eight — and the reason is instructive: a refinement derived from four in-sample populations turned a seven-of-eight prediction into a four-of-eight one.**



***Index Terms*—algorithmic fairness, levelling down, demographic parity, in-processing, selection rate, disparate impact**


## I. Introduction

A fairness constraint says two groups' rates should be close. It says nothing about how. The gap can close because the disadvantaged group is lifted, or because the advantaged group is pulled down. Both satisfy the constraint; only one is what anybody meant.

That this ambiguity exists is established. Mittelstadt *et al.* [2] made levelling down the centre of a well-known critique; Maheshwari *et al.* [4] report that it "often goes unnoticed in the overall performance of the model"; Ferry *et al.* [3] built an auditing framework because an aggregate score reports neither how many people were affected nor in which direction. We take all of that as given.

Our question is the one this literature leaves open: **when?** A deploying team cannot act on "this sometimes happens". It can act on a rule that says which way its own system will move, before it moves.

### A. Scope, stated first

This is a paper about **attribute-blind in-processing** constraints — the regime in which the protected attribute is unavailable at decision time. That is where almost every deployed system sits, because per-group thresholds constitute disparate treatment on their face in most jurisdictions. Under post-processing, which does read the attribute, the effect we report disappears (Section VII). This is not a weakness of the finding but the boundary theory draws, and we verify it from both sides.

### B. Contributions

1) **An empirical characterisation of a conditionality that theory predicts.** Contemporaneous theory [5] establishes that in the attribute-blind regime the direction is distribution-dependent. It contains no experiments and its conditions are ordering relations on score regions rather than measurable quantities. We supply the measurement across 18 populations and five domains, and find its stated conditions hold on **0 of 26** arms.
2) **Separation of the predictor from its confound.** Varying the selection rate by moving a label threshold cannot distinguish "the rate predicts the direction" from "task difficulty does". Holding the task exactly fixed and moving only the decision rule separates them. We claim a *predictive* relationship within a population under the tested regime, not a causal mechanism: no pooled slope transfers across populations (Section VIII).

3) **A procedure, not an observation.** The same design lets a practitioner locate their own crossover on a deployed model. Four independent estimates of the mortgage crossover agree.
4) **Boundaries stated rather than discovered by a referee**, including a computable test for when the procedure cannot be used at all.

## II. Related Work

**Whether levelling down is universal.** A 2026 theory paper [5] proves in a population-level Bayes framework that the answer depends on the deployment regime. Where the protected attribute is available at decision time, fairness "necessarily (weakly) improves outcomes for the disadvantaged group and (weakly) worsens outcomes for the advantaged group". Where it is not — the attribute-blind regime, which is ours — "the impact of fairness is distribution-dependent: fairness can benefit or harm either group . . . leading to either leveling up or leveling down." **We therefore do not claim the conditionality itself.**

**Levelling down, and the remedy.** Mittelstadt *et al.* [2] argue that fairness interventions frequently equalise by degrading the better-off group, and *also propose the remedy*: minimum rate constraints requiring "that every group has, at least, a minimal selection rate, precision, or recall". We claim neither the observation nor the remedy as novel. Our contribution relative to that work is narrow and checkable: a scope condition they do not have; in-processing rather than post-processing, so no model reads the attribute when predicting; a population-level floor stacked *with* parity rather than a per-group floor replacing it; and held-out replication where they report training-set results.

**Auditing individual impact.** Ferry *et al.* [3] propose a framework whose first three dimensions are impact size, change direction and decision rates. That is the decomposition we use throughout, and we claim no part of it. What we add is what those axes report at scale: the change-direction dimension is not a property of the method being audited but of the population it is applied to.

**The shape of the optimal fair classifier.** Corbett-Davies *et al.* [10] show that for several fairness definitions the optimal constrained classifier takes the form of group-specific risk thresholds. This gives a *bound* rather than another baseline: it predicts that levelling down is not an artifact of the reduction's search, because the optimal constrained classifier withdraws in the same way.

**Remedies that avoid harm.** Minimax group fairness [11] minimises the worst group's error rather than equalising anything, and is the natural alternative to a parity constraint for a team whose concern is harm rather than equality. We benchmark against it rather than only against the unconstrained reduction.

**The selection-rate axis.** The nearest prior work is Goethals *et al.* [8], which studies the cost of fairness as a function of available budget and reports that "the level of available resources significantly influences this cost, a factor overlooked in previous evaluations". The difference is structural: in their setting positive decisions are a fixed resource, so the total is constant by construction and the directional effect we measure cannot arise.

**Fairness gerrymandering and dataset monoculture.** Kearns *et al.* [6] show constraints on marginal groups can be satisfied while subgroups are badly treated. Ding *et al.* [7] argue the field should stop drawing conclusions from UCI Adult and supply ACS-derived replacements. Notably they also *build the knob this paper turns*: the $50,000 threshold "was chosen so that this dataset can serve as a replacement to UCI Adult, but we also offer datasets with other income cutoffs". They do not connect it to the direction of a fairness intervention — their text contains no occurrence of "selection rate", "acceptance rate" or "leveling down" — so the instrument existed and the question was not asked of it. We replicate across their populations and then find that is *not sufficient*, because they share one survey instrument.

## III. Setup and Method

**Data.** Eighteen independent populations across four instruments: UCI Adult (45,222 rows); ACS Income [7] across twelve US states and two protected attributes; HMDA mortgage records for two states, by race and sex and split by loan purpose; COMPAS [9] (5,278 rows on the race arm after ProPublica's documented filter); LSAC bar passage (20,798 rows); and the Dutch national census of 2001 (60,420 rows).

The last three were added to leave the instruments the finding was formed in. The Dutch census carries a between-group gap of 0.298, roughly twice Adult's, making it the test of the standard alternative explanation. LSAC was chosen because bar passage is naturally generous (base rate 0.890).

**Convention.** $y = 1$ is always the favourable outcome. This required inverting COMPAS's recorded target, which encodes recidivism, and declaring *Female* the advantaged group on its sex arm, since women in that cohort reoffend less. Both are enforced by automated checks, because either error produces every directional result sign-inverted without raising.

**Mitigation.** `ExponentiatedGradient` [1] at $\varepsilon = 0.01$ unless stated. No model reads the protected attribute at prediction time except where post-processing is explicitly under test.

**Exclusion rules.** An arm is discarded if the baseline demographic-parity gap is below 0.05, or if the baseline classifier's accuracy falls below $\max(p, 1-p)$ — the accuracy of always predicting the more common label. The second rule matters more than it sounds (Section X).

## IV. Parity Is Bought by Withdrawal — Where It Is

Table I gives the ablation. Every method reaches its target: demographic parity falls from 0.186 to as little as 0.015 for under two accuracy points, on a linear model and on an adversarial one alike. This is the part that works, and it is background rather than contribution.

**Every one of them shrinks the pool**, by 7.9% to 22.1%. Not one closes the gap primarily by extending decisions to the

TABLE I
SIX IN-PROCESSING METHODS ON UCI ADULT, FIVE SEEDS. EVERY ONE REACHES PARITY, AND EVERY ONE SHRINKS THE POOL.

| Method | Acc. | DP | EO | DI | Δpool |
|---|---|---|---|---|---|
| Unconstrained | 0.847 | 0.186 | 0.095 | 0.300 | — |
| ExpGrad (DP) | 0.828 | **0.018** | 0.280 | 0.895 | −20.5% |
| GridSearch (DP) | 0.827 | **0.015** | 0.304 | 0.986 | −22.1% |
| Adversarial deb. | 0.826 | 0.020 | 0.250 | 0.889 | −14.8% |
| Prejudice remover | 0.837 | 0.065 | 0.188 | 0.686 | −10.2% |
| ExpGrad (EO) | 0.836 | 0.107 | **0.032** | 0.521 | −7.9% |

TABLE II
WHO PAYS, UCI ADULT. THE RATE-LEVEL VIEW AND THE PEOPLE-LEVEL VIEW OF THE SAME INTERVENTION DISAGREE SYSTEMATICALLY.

| Method | Share (rates) | Share (people) | Exchange |
|---|---|---|---|
| ExpGrad (DP) | 0.575 | **0.742** | 2.68 |
| GridSearch (DP) | 0.575 | **0.743** | 2.69 |
| Adversarial deb. | 0.508 | **0.700** | 1.96 |
| Prejudice remover | 0.498 | **0.673** | 2.04 |
| ExpGrad (EO) | 0.531 | **0.660** | 1.92 |

disadvantaged group. Eighteen of nineteen survey arms do the same.

### A. Rates understate what is happening to people

The decomposition of [3] splits the closed gap into the part paid by the advantaged group losing ground and the part gained by the disadvantaged group. Measured in *rates*, that split is close to even — 0.50 to 0.58 of the closure comes from withdrawal (Table II).

Measured in *people* it is not. Between 0.66 and 0.74 of the individuals whose decision changed are members of the advantaged group losing a favourable one. Equal-looking rate changes fall on groups of unequal size, so the rate-level view — which is the view a fairness dashboard reports — understates the burden counted in individuals in every method tested. The divergence between the two views is itself predictable, from a cross-flow diagnostic, at $r = +0.885$ across populations.

The **exchange rate** makes the same point without a share: ExpGrad under demographic parity destroys 2.68 favourable decisions for every one it creates.

**And then it reverses.** On HMDA mortgage data the identical constraint *grows* the pool by 4.3%, at an exchange rate of 0.50 — two favourable decisions created for every one destroyed. The parity metric reports 0.018 on the population that destroyed a fifth of its favourable decisions, and 0.010 on the one that created more than it destroyed.

## V. WHAT PREDICTS THE DIRECTION

### A. A single-factor design

ACS Income's label is "earns more than \$50,000" — a benchmark convention, not a fact. Moving it on one fixed state varies the selection rate while holding population, instrument, feature set, group ratio and proxy structure fixed. The

TABLE III
THE RELATIONSHIP ACROSS INSTRUMENTS

| Instrument | Domain | $r$ |
|---|---|---|
| ACS Income | income prediction | +0.801 |
| HMDA | mortgage lending | +0.858 |
| COMPAS | criminal justice | +0.870 |
| Dutch census | occupational status | +0.915 |
| LSAC | legal education | unsweepable |

change in the pool rises with the baseline selection rate at $r = +0.801$, and $+0.980$ once the confounded between-group gap is partialled out. The sign flips: 22 decisions destroyed per one created at a rate of 0.03, against 0.75 at 0.89.

### B. The transition appears without being manufactured

Pooling two states across five real loan products — nothing manipulated — home improvement at a selection rate of 0.555 levels *down* (−1.57%) while refinancing at 0.871 levels *up* (+2.95%), with $r = +0.803$ and Spearman $\rho = +0.900$. A bank running one constraint across its loan book would take opportunities away in one product and hand them out in another, with its fairness report showing success in both.

### C. It is the rate, not the difficulty of the task

The design above moves the *label*. "Earns over \$100,000" is not a rarer version of "earns over \$20,000"; it is a harder problem. That design therefore cannot separate two explanations which predict everything observed so far.

We separate them by holding the task completely fixed — same rows, features, groups and *fitted scores* — and moving only where the model draws its line. The relationship reproduces, and so does the crossover's location: on Oregon the two routes give 0.362–0.653 and 0.353–0.637. **Task difficulty is not doing the work.**

With twelve points chosen from each population's viable band, the relationship holds on the Dutch census (+0.946), South Carolina (+0.905) and COMPAS (+0.844). Alabama and Kentucky return *negative* values on eleven and ten arms; their viable bands top out at 0.566 and 0.561, at or below every crossover measured, so both sweeps lie almost entirely on one side of the transition. **We therefore claim the relationship across the crossover and withdraw any claim that it holds within the low-rate region alone.**

### D. Five domains, and the group-gap alternative

Fig. 1 shows every arm, and Table III the summary. The pre-registered alternative — "this is a property of income prediction and American mortgage lending", requiring $|r| < 0.30$ — loses on both new instruments.

The oldest competing account is that the *gap between groups* drives the direction. The Dutch census has a gap of 0.298, roughly twice Adult's, making it the population where that explanation should win. It gives $r = +0.915$ across all six arms, none excluded.

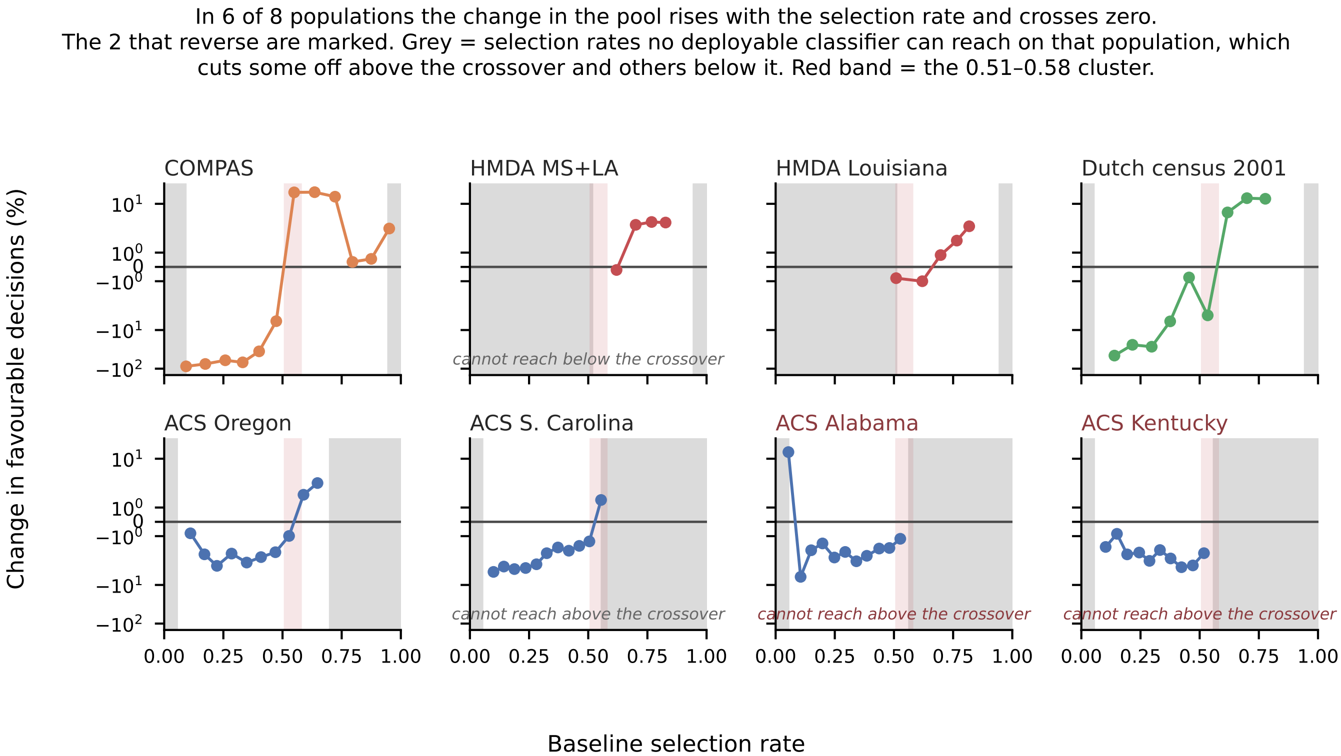


Fig. 1. Every retained arm from every densely swept population. Within a population the change in the pool rises with the baseline selection rate and crosses zero; the two that reverse are marked, and both are sweeps whose viable band confines them below the crossover. Panels are deliberately *not* pooled: across populations a single slope reaches only $r = +0.487$ and fails its pre-registered bar, so one scatter would assert a common curve the data rejects. Vertical band: the 0.511–0.576 cluster of Table VI.

TABLE IV
WHAT EACH MANIPULATION PRESERVES

| **Manipulation** | **Direction** | **Magnitude** |
|---|---|---|
| Route to the rate | unchanged | differs 3× |
| Tolerance, 25× range | unchanged | differs 4× |
| Boosted trees (5/5 pops.) | unchanged | differs 3× |
| Attribute, sex → race | unchanged | — |
| Criterion → equalized odds | weaker, unevenly | differs 8× |
| Optimiser → post-proc. | **vanishes** | — |

### *E. Robustness*

Table IV summarises. **Within a population, and in the attribute-blind regime, the selection rate predicts which way and orders how much. Neither transfers between populations**: the signed distance from a population's own crossover gives $\rho$ of $+0.96$, $+0.95$ and $+0.78$ in three of four populations, but a pooled slope reaches only $r = +0.487$ against a pre-registered $+0.70$ bar.

## VI. A Constraint on One Attribute Leaves Subgroups Behind

A constraint is imposed on the attribute someone chose to name. Kearns *et al.* [6] showed that marginal constraints can be satisfied while structured subgroups are badly treated. Table V is that failure mode measured on Adult.

TABLE V
UCI ADULT. CONSTRAINING SEX TO 0.020 LEAVES THE WORST *Sex×Race* SUBGROUP GAP AT 0.178 — NINE TIMES LARGER — AND MOVES WHICH SUBGROUP IS WORST OFF.

| **Arm** | **Sex** | **Subgroup** | **Worst off** |
|---|---|---|---|
| Unconstrained | 0.190 | 0.315 | F×Black |
| Constrained on sex | **0.020** | **0.178** | M×Black |
| Constrained on both | 0.016 | 0.048 | F×White |

Constraining sex drives its gap from 0.190 to 0.020 — a success by the certifying metric. The worst *Sex×Race* subgroup gap falls only from 0.315 to 0.178, which is **nine times** the gap that was constrained. The subgroup carrying the residual also *changes*, from Female × Black to Male × Black, so an auditor tracking the originally worst-off cell would see improvement while a different cell absorbed the cost.

This replicates across populations and is more severe than Adult suggests: 13.2× in Mississippi. It is bounded, however, by how large the minority is — the effect scales with minority share at $r = -0.671$, and does not appear at all below a threshold. We present this as an empirical replication of [6] and [4] with a boundary attached, not as a new observation.

**A measurement hazard worth naming.** Roughly five

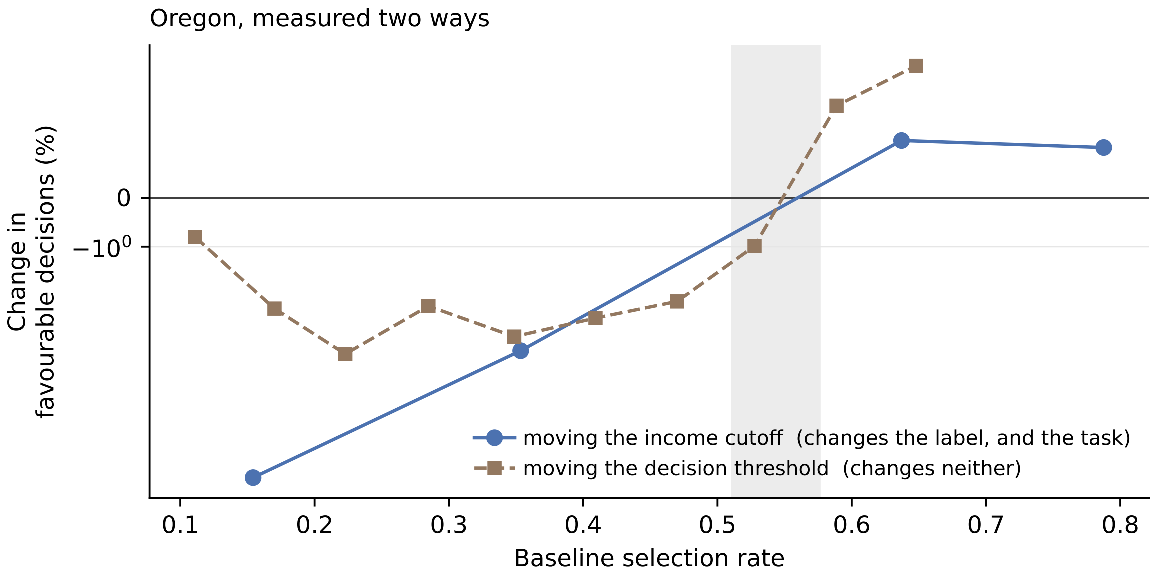


Fig. 2. The design that separates the selection rate from the difficulty of the task. Moving the income cutoff changes the label, and therefore how hard the problem is; moving the decision threshold changes neither, only where the fitted model draws its line. Both routes cross zero inside the same band. If difficulty were doing the work, the second curve would not reproduce the first.

subgroups per population are too small to support any estimate. A rate with an empty denominator must return undefined rather than zero, or an unmeasurable subgroup masquerades as a perfectly fair one — which is the failure this analysis is most likely to produce and hardest to notice.

## VII. The Regime Boundary

The last row of Table IV is not fragility. Post-processing reads the protected attribute at prediction time; the reduction does not. That is the distinction on which the theory of [5] is built.

Running post-processing at each population's natural operating point across 17 populations gives $r = -0.024$, against $+0.585$ for the reduction on the same populations. The pre-registered prediction failed and its naive alternative won outright.

**In the attribute-aware regime the theory says there is nothing to predict**, because the direction is determined. Its prediction for that regime is that the advantaged group loses and the disadvantaged gains, always. On our populations that holds in **18 of 18**, without exception. That check was post-hoc, so we repeated it as a confirmatory test on nine populations never previously post-processed, with the bar set at all nine and committed before the runs: it holds **9 of 9**. The regime result therefore stands at 27 of 27, nine of them predicted in advance.

Section X notes that the same theory's *conditions* hold on 0 of 26 arms. Its regime distinction holds on 18 of 18. The theory is right about which regime you are in mattering, and wrong about which quantity decides it within one — which is what an empirical study can settle and a population-level argument cannot.

TABLE VI
Crossover location with bootstrap intervals over seeds (2,000 resamples). "Located" is the fraction of resamples in which a sign change could be bracketed at all.

| Population | Domain | Crossover (95% CI) | Loc. |
|---|---|---|---|
| *Non-lending* | | | |
| COMPAS | crim. justice | 0.511 (0.434–0.524) | 83% |
| S. Carolina | income | 0.530 (0.526–0.533) | 99% |
| Oregon | income | 0.558 (0.556–0.560) | **77%** |
| Dutch census | occupation | 0.576 (0.572–0.580) | 100% |
| *Lending — one market, three overlapping estimates* | | | |
| HMDA pooled | mortgages | 0.660 | — |
| HMDA Louisiana | mortgages | 0.659 | — |
| HMDA refinance | mortgages | 0.758 | — |

## VIII. Boundaries

### A. The crossover is nearly stable

Table VI separates the two groups deliberately, because averaging them would hide the only disagreement in the data. Four non-lending populations, across three domains and two countries, have overlapping intervals. **Carrying those intervals the cluster spans roughly 0.43–0.58**, not the 0.511–0.576 that point estimates alone suggest: COMPAS's interval is six times wider than the Dutch census's, and Oregon's bracket fails to exist in 23% of seed resamples.

Every lending estimate sits above every non-lending one. They come from a *single* market measured three overlapping ways, so "lending is different" is a hypothesis here and not a finding.

**We therefore treat a crossover near 0.54 as an empirical prior for auditing, not a theoretically justified constant.**

TABLE VII
SENSITIVITY TO THE PARITY-GAP EXCLUSION. CORRELATION PER POPULATION AS THE THRESHOLD IS LOOSENED.

| Population | 0.01 | 0.02 | 0.05 | 0.10 |
|---|---|---|---|---|
| Dutch census | +0.851 | +0.908 | +0.946 | +0.938 |
| COMPAS | +0.844 | +0.844 | +0.844 | +0.871 |
| S. Carolina | +0.012 | +0.012 | +0.905 | +0.849 |
| Oregon | +0.095 | +0.095 | +0.664 | — |
| Alabama | −0.368 | −0.368 | −0.368 | +0.296 |
| Kentucky | −0.590 | −0.590 | −0.654 | — |

Nothing in this work explains why 0.54 rather than some other value, and Section X records an attempt to derive it that failed.

Two population properties appeared to explain the residual, at $r = +0.724$ and $+0.865$. They are collinear with each other at $+0.947$ and neither is distinguishable from chance at four populations ($p = 0.277$, $p = 0.135$). We report them as a hypothesis for a larger study, not a formula.

### *B. The procedure is unusable on very generous tasks*

Varying the selection rate by moving a threshold works by *degrading* the classifier. On a task where most people already qualify, reaching a low selection rate requires a model that refuses qualified people, and past a point that model is worse than a constant. Defining the *viable band* as the range of selection rates reachable by a classifier still beating $\max(p, 1-p)$: Dutch spans 0.895, COMPAS 0.859, typical ACS $\approx 0.51$, and LSAC only **0.056**. On LSAC, where 89% pass the bar, no choice of thresholds produces a usable sweep. This is computable before running anything.

When the viable band forbids a sweep, the crossover can still be located by comparing natural sub-populations — which is how the mortgage crossover was first found.

### *C. A sealed prediction, and what it cost*

Every result above is a correlation computed after the arms were run, which is the weakest support for a claim of the form *you can know in advance*. We therefore ran the strongest format available: nine populations whose direction had never been measured, with the rule, the populations and the pass mark committed before any of them existed.

**It failed. Four of eight, against a bar of seven, not beating a constant.**

**What failed is a refinement, not the rule.** Table VII shows the exclusion threshold doing real work: South Carolina moves from $+0.905$ to $+0.012$ when it is loosened. Investigating that, we found the newly admitted arms are not noise — at twenty seeds they are consistently *positive*, $+8.73$, $+6.85$, $+9.09$, six standard errors from zero — and concluded the relationship turns back up below a selection rate of 0.10. The sealed rule carried that refinement.

All three held-out low-rate arms came out strongly *negative*. Scoring the same eight arms under the simpler rule we held before the refinement — down below the crossover, up above it — gives **seven of eight**, its only miss being Maryland at 0.508, within 0.03 of the crossover on an effect of $+0.78\%$.

TABLE VIII
THE SEALED PREDICTION. POPULATIONS NEVER PREVIOUSLY MEASURED; PREDICTIONS COMMITTED BEFORE THE RUNS.

| Population | Rate | $\Delta$pool | Pred. | Actual |
|---|---|---|---|---|
| MO, \$100k | 0.026 | −16.48% | up | **down** |
| AZ, \$100k | 0.037 | −23.34% | up | **down** |
| IN, \$70k | 0.081 | −23.32% | up | **down** |
| TN, \$50k | 0.255 | −1.68% | down | down |
| MD, \$50k | 0.508 | +0.78% | down | **up** |
| MI, \$30k | 0.612 | +0.85% | up | up |
| NC, \$20k | 0.789 | +0.16% | up | up |
| GA, \$10k | 0.905 | +0.30% | up | up |
| Taiwan 2005 | 0.885 | +0.68% | up | up |

That seven is post-hoc and we do not claim it; what was sealed scored four.

**The refinement was route-specific.** Its arms reach a rate near 0.05 by thresholding one fixed score vector; the sealed arms reach it with a genuinely rarer label. At that rate the two disagree completely, $+8.73$, $+6.85$, $+9.09$ against $-16.48$, $-23.34$, $-23.32$. So Section V-C qualifies: the two routes agree through the middle of the range and diverge at the bottom, and the accuracy rule does not catch the divergent arms — they clear the trivial-predictor bar and remain artifacts.

A refinement drawn from four in-sample populations, internally consistent and backed by a twenty-seed replication, made the prediction worse than the rule it replaced and no better than a constant. Nothing in the in-sample data could have shown that.

### *D. Small populations*

### *E. Small populations*

Below roughly 2,500 test subjects the method's own randomness exceeds the entire effect of the constraint. This is a specific instance of predictive multiplicity [12] — models with equivalent aggregate performance disagreeing substantially on individuals — and we position it as a caution supported by that literature rather than as a novel finding. COMPAS has 1,584 and flips sign between seeds on two of four arms; LSAC has 6,240 and flips on none. COMPAS carries direction here and never magnitude.

## IX. WHAT TO DO ABOUT IT

Algorithm 1 states the procedure. Three of its steps are worth drawing out, because each is a place where a plausible-looking run returns a meaningless answer.

**It can refuse.** Lines 6–8 compute the viable band and exit before any constraint is fitted, if no threshold reaches a low selection rate with a model still beating the trivial predictor. On LSAC that band spans 0.056 and the procedure declines, rather than returning arms that would be discarded later.

**It can return VOID.** Line 16 requires the pool to have moved at all. A correlation computed over arms that barely move is fitted to noise: Connecticut returns $r = -0.924$ on a spread of a tenth of a percentage point, and without this test that number reads as a refutation of the whole finding.

**Algorithm 1** Locate the crossover; decide whether to add a floor

**Require:** deployed model $h$, held-out $(X, y, a)$, fairness constraint $C$
**Ensure:** predicted direction, and whether a selection-rate floor is needed

1: **if** $C$ does not constrain selection rates **then**
2:     **return** OUT OF SCOPE {e.g. equalized odds}
3: **end if**
4: $\pi \leftarrow \max(\bar{y},\, 1 - \bar{y})$ {trivial-predictor accuracy}
5: $s \leftarrow h(X)$
6: $\mathcal{V} \leftarrow \{\tau : \mathrm{acc}(s > \tau) \geq \pi \ \wedge\ 0.05 \leq \overline{(s > \tau)} \leq 0.95\}$
7: **if** $\mathrm{span}(\mathcal{V}) < 0.40$ **then**
8:     **return** UNSWEEPABLE; compare natural subpopulations
9: **end if**
10: $T \leftarrow 12$ thresholds from $\mathcal{V}$, spread evenly in *selection rate*
11: **for all** $\tau \in T$ **do**
12:     $r_\tau \leftarrow \overline{(s > \tau)}$; fit $C$ on the arm thresholded at $\tau$
13:     $\Delta_\tau \leftarrow$ % change in favourable decisions
14:     discard $\tau$ if parity gap $< 0.05$ or $\mathrm{acc} < \pi$
15: **end for**
16: **if** $\max \Delta - \min \Delta < 2$ points **then**
17:     **return** VOID {nothing moved; not a refutation}
18: **end if**
19: $c \leftarrow \big(\max\{r_\tau : \Delta_\tau < 0\},\ \min\{r_\tau : \Delta_\tau > 0\}\big)$
20: **if** $c$ is empty **or** $c$ is far from $0.54$ **then**
21:     inspect the sweep before believing it
22: **end if**
23: $r_0 \leftarrow$ selection rate of the deployed model
24: **if** $r_0 < \min c$ **then**
25:     **return** WITHDRAWAL; add a selection-rate floor
26: **else**
27:     **return** EXTENSION; the floor buys little
28: **end if**

**It returns a sign, never a size.** The distance from the crossover *orders* the effect within a population, so a team can rank its own segments; but no slope transfers between populations, so lines 25 and 27 return a direction and stop there.

The floor is **not a new method**: a selection-rate floor is a linear moment constraint inside the framework of [1], and a variant of the minimum rate constraints of [2]. It costs about 0.12 accuracy points and moves the exchange rate from 1.47 favourable decisions destroyed per one created to 0.88 across nineteen arms, landing below 1.0 in sixteen of them, with the benefit scaling with the damage at $r \approx -0.99$.

## X. Method, and What We Got Wrong

Every prediction was registered with its thresholds *and the naive alternative it had to beat*, before the data existed. Three episodes matter for reading the results above.

**A held-out prediction, scored as a failure.** The clearest evidence format available is a prediction registered before the data exists. We ran one: after adding the accuracy rule below, four HMDA populations the sweep had never touched were swept and scored under it, with the prediction that each would show the relationship, clear the spread guard and bracket a crossover. **Two of four did**, so the test is scored a failure. Its named alternative — that lending is a domain where this route does not work — nonetheless loses on all four, with $r \geq +0.80$ throughout and $+0.995$ on refinance. Three independent estimates of the lending crossover agree (Table VI).

**A pre-registered test that failed.** The rule does not survive equalized odds at the bar we set: $r = +0.644$ on two states, $+0.334$ on five, against a $+0.70$ threshold. Alabama alone would have given $+0.822$ with every prediction holding.

**Two headline results withdrawn.** The operating-point design varies the selection rate by degrading the classifier, and nothing bounded how far. Excluding arms beaten by the trivial predictor leaves Alabama with two arms and LSAC with one: $r = +0.979$ and $r = +0.968$, both previously reported, are void. The same exclusion *rescued* mortgage lending, which had failed outright.

**A derivation that a constant beat.** The most obvious extension of this work is a theoretical account of *why* the selection rate should matter. A general theory already exists and we cite it: [5]'s Theorem 3 gives explicit conditions, over the extrema of the score-region quantity $\zeta(x) = (\eta(x) - c)/\nu(x)$, under which every group's rate weakly falls or weakly rises. Those conditions are ordering relations rather than anything a practitioner can compute, and Section VII reports that they hold on **0 of 26** of our arms.

We attempted the missing step — a derivation predicting the sign from the selection rate directly — and pre-registered its bars. **It cleared them and was then beaten by a constant**, 12 correct against 13 out of 14 held-out arms. That failure is why this paper reports an empirical regularity rather than a mechanism, and why we do not attempt a fourth derivation here.

**A plausible result that was arithmetic.** A post-processing comparison appeared to show a clean reversal. It was void: the method re-derives its own thresholds and returned an identical model at all six operating points, so the measured effect was a constant divided by a shrinking baseline. What exposed it was an exchange rate of 0.000.

## XI. Limitations

**The rule has not passed a sealed out-of-sample test.** Section VIII-C reports one that failed at four of eight. A simpler form scores seven of eight on the same arms, but that comparison is post-hoc; the honest position is that the predictive claim is supported within populations and *not yet* demonstrated on unseen ones.

**The two manipulations diverge at low selection rates.** The operating-point route and the label route disagree completely below a rate of about 0.10, and the accuracy rule does not separate them. We claim the relationship across the transition only.

**Magnitude does not transfer** between populations, only within them. **The crossover clusters but is not explained**: two candidate predictors are collinear at $+0.947$ and neither survives four populations. **Coverage** is five instruments, three countries and two decades; Taiwan is the only non-Western population, and one population is not coverage. **The regime result** was post-hoc at 18 of 18 and has since been confirmed pre-registered at 9 of 9, but it confirms a published theorem rather than establishing one. **The remedy is a variant**, not a discovery.

## XII. Discussion

The finding a practitioner can use is small and specific: **look up the rate at which your model currently says yes, locate your own crossover, and you can anticipate whether a parity constraint is about to give or to take.** We do not claim the selection rate *causes* the direction; we claim it predicts it, within a population, in the attribute-blind regime, and with the boundaries of Section VIII attached.

What makes that worth saying is not that it is surprising once stated. It is that the certifying metric — the number a deployment team reports, an auditor checks, and a regulator may one day require — is identical in both worlds. A constraint that withdraws a fifth of all favourable decisions and one that creates more than it destroys produce the same certificate.

The literature already knew the metric was silent. What was missing was a way to know, in advance and from something a team already has, which of the two silences you are standing in.

## Reproducibility

Every load-bearing figure is re-derived from stored results by an automated check that fails if a document and its data disagree; 28 such checks accompany this work. Pre-registrations are committed before the runs they govern.

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